

Maximum Mark: 40

Final Sem Exam

Time: 3 hours

1. (20 points) [4+4+2+2+6+2]

(a) (4 points) Draw the complete energy level representation of the hyperfine splitting of the Deuterium atom (2_1D) for $n=2$ level (nuclear spin quantum number, $I=1$). The diagram should start with the non-relativistic (Schrodinger equation) energy level and should show the effect of all the interactions (Fine Structure, Lamb shift, etc). You need not worry about the exact scale but the relative ordering of the levels should show the true representation.

(b) (4 points) Consider an H atom of which the wave function at $t=0$ is the following superposition of energy eigenfunctions $\psi_{nlm}(\vec{r})$

$$\Psi(\vec{r}, t=0) = \frac{1}{\sqrt{14}} [3\psi_{100}(\vec{r}) - 2\psi_{200}(\vec{r}) + \psi_{322}(\vec{r})]$$

(i) Is the wave function an eigenfunction of the parity operator?

(ii) What is the probability of finding the system in the ground state (100)? In the state (200)? In the state (322)? In another eigenstate?

(iii) What is the expectation value of the energy? Of the operator $\hat{L}^2$? Of the operator L_z ? in general

(c) (2 points) Estimate the ratio of typical vibrational (nuclear) and rotational energy of molecules with respect to its electronic energy. For the latter, you may assume the simplest molecular structure i.e. diatomic molecule.

(d) (2 points) Show that $M_{ba} = -M_{ab}^*$ where $M_{ba} = \langle \psi_b | e^{i\vec{k} \cdot \vec{r}} \vec{e} \cdot \vec{\nabla} | \psi_a \rangle$ (Matrix related to radiative transition; show all steps and assumptions clearly).

(e) (6 points) Assuming that L-S coupling holds, list all the possible terms $^{2S+1}L_J$ for the electronic configuration $(nd^3) n p^2$ and $n p \ n' p' \ n' d$. Briefly states the reason for these terms.

(f) (2 points) Draw the general curve for electronic energy $E_e(R)$ (or atomic potential $E_a(R)$) for a bound state of diatomic molecule with proper labels where R is internuclear separation. Briefly state the reason for the particular shape.

(g) (3 points) Derive the general Einstein's A, B coefficients and the relation between them for a two energy levels system (E_a , E_b) with degeneracy g_a and g_b respectively.

(h) (4 points) Show that the energy corrections after including fine structure effect (relativistic) to the H-like atoms agree with the exact solution (given below; solution of the Dirac-equation) through order $(Z\alpha)^2$

$$E_{nj}^{exact} = mc^2 \left\{ \left[1 + \left(\frac{Z\alpha}{n - j - 1/2 + \sqrt{(j + 1/2)^2 - Z^2\alpha^2}} \right)^2 \right]^{-0.5} - 1 \right\}$$

Show/list all your steps and assumptions clearly.

5. (5 points) For a weak external magnetic field, B_0 in the z-direction, draw the splitting of the energy level of an H-atom and mark all the possible transitions under the dipole transition rule between the $n=2$ and $n=1$ levels.

5. (8 points) Construct all the spin wave functions (χ_{S,M_S}) for a 3-electron system (Li atom) and write the corresponding S and M_S values. Briefly list the approach. Does these spin wave functions have symmetry as in the 2-electron system case?

d.u.

Useful Results

- $h = 6.63 \times 10^{-34} \text{ J s}$, $c = 3 \times 10^8 \text{ m/s}$, $e = 1.6 \times 10^{-19} \text{ C}$
 $m_e = 9.1 \times 10^{-31} \text{ kg}$, $m_p = 1.673 \times 10^{-27} \text{ kg}$, $m_p = 938.3 \text{ MeV}$
 $m_n = 1.675 \times 10^{-27} \text{ kg}$, $m_n = 938.6 \text{ MeV}$
- Fine structure splitting, $\Delta E_{nj} = E_n^0 \frac{(Z\alpha)^2}{n^2} \left(\frac{n}{j+1/2} - \frac{3}{4} \right)$;
- $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$;
- $E_n^0 = -\frac{1}{2}\mu c^2 \frac{(Z\alpha)^2}{n^2}$
- Zeeman shift in weak magnetic field, $\Delta E_{m_j} = \underline{g_j \mu_B B_0 m_j}$ $j = l + s$

$$g_j = 1 + \frac{j(j+1)+s(s+1)-l(l+1)}{2j(j+1)}$$